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Post-movement beta synchronization. A correlate of an idling motor area?

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Abstract

Post-movement beta (around 20 Hz) synchronization was investigated in 2 experiments with self-paced finger extension and flexion and externally paced wrist movement. The electrodes were fixed over the sensorimotor area in distances of 2.5 cm. It was found that after a brisk finger movement the desynchronized beta rhythm displayed a fast recovery and a short-lasting synchronization within 1 sec. This post-movement beta synchronization was maximal over the contralateral hemisphere and localized slightly more anterior to the maximal desynchronization of the hand area mu rhythm. The post-movement beta synchronization is interpreted as a correlate of "idling" motor cortex neurons.

Keywords: Finger movement; Beta rhythm; Event-related synchronization; Idling area

1. Introduction

The mu and central beta rhythms are associated with motor behavior and, therefore, of special interest because the brain is organized primarily to plan and control motor behavior (Yingling, 1980). As concerns beta rhythms, these have been investigated since the pioneer work of Berger (1930) and Tönnies (1934) who first recorded electrocortical beta band activity (16–22 Hz) in the precentral cortex. Later, Jasper and Andrew (1938) reported on precentral beta rhythms at an average frequency of about 25 Hz, depressed by tactile stimulation of the contralateral side of the body and not affected by visual stimulation. Jasper and Penfield (1949) found that the central fissure represents a most important borderline separating beta and alpha rhythms. The most characteristic feature of the precentral beta rhythm was its blocking or desynchronization localized over the motor hand area by fist clenching. Jasper and Penfield (1949) interpreted the precentral beta rhythm as the idling activity of that region of resting motor cortex from which an EEG is derived.

A more recent study (Papakostopoulos et al., 1980) showed beta activity between 16 and 32 Hz in pre- and postcentral areas in man. A low amplitude rhythm from central locations between 12 and 15 Hz termed "sensorimotor rhythm" (SMR) was reported in man by Sterman (1977). An analogous rhythm can be found in immobile cats (Howe and Sterman, 1972). Sensorimotor rhythms of 12–14 Hz in mobile and immobile cats were reported by Chase and Harper (1971). They suggested that rhythmic 12–14 Hz activity in the sensorimotor cortex is a correlate of localized sites of "idling" cortical areas. A similar synchronized 12–16 Hz EEG activity was reported over the sensorimotor cortex during inhibition of motor behavior in cats (Roth et al., 1967). Rougeul et al. (1979) studied fast somato-parietal rhythms around 18 Hz in monkeys and found that these rhythms are distributed over the SI hand area and over the posterior parietal cortex and that they are blocked by the smallest body movement.

Beta desynchronization during voluntary hand movement was quantified and displayed in the form of event-related desynchronization (ERD) time courses by Pfurtscheller (1981). This beta ERD occurred in parallel with the mu ERD. A movement-related desynchronization in the lower beta frequency band recorded over the sensorimotor hand area was reported not only in a self-paced

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index finger movement task but also with foot movement (Toro et al., 1994). One recent neuromagnetic study of mu and beta rhythms in man reported a desynchronization followed by a fast rebound of 20 Hz beta oscillations in response to thumb movement (Salmelin and Hari, 1994).

Central beta band rhythms display, on the one side, a desynchronization (ERD) prior and during movement and on the other side, a rebound in the form of a phasic event-related synchronization (ERS) after movement (Stancák and Pfurtscheller, 1995). This paper is focused on this post-movement beta ERS during self-paced finger movement (experiment 1) and reports on data on the spatial location and temporal pattern of this beta ERS over pre- and postcentral areas in man.

Beside EEG data from self-paced finger movements (experiment 1) data from externally paced wrist movements (experiment 2) are also reported. Experiment 2 was designed to help constructing an EEG-based Brain-Computer Interface (BCI; Pfurtscheller et al., 1993a) where a subject is trained to move a cursor on a monitor through analysis and classification of EEG signals from sensorimotor areas during wrist movements. A detailed description of this study can be found elsewhere (Pfurtscheller et al., 1996).

2. Methods and subjects

2.1. Experiment 1

2.1.1. Subjects

Ten healthy subjects (5 females and 5 males, mean age 26.1 years, S.D. = 4.0). volunteered in this study. They were graduate and postgraduate students and were paid for their participation. All subjects, except one, were right-handers according to self-report.

2.1.2. Procedure

Subjects were informed about the purpose and procedure of the experiment, and electrodes were attached. They were sitting in a semi-reclining armchair with their hands resting on platforms in a darkened room with their eyes closed. The subjects were given a 5 min training period to perform self-paced movements in intervals not shorter than 13 sec and not longer than 17 sec. It was explained to the subjects how to perform brisk movements. A brisk movement consisted of rapid extension and flexion of the right index finger at an angle of 20–30°. Durations of finger movements were checked on the computer screen. One series of brisk movements comprised 90–100 movements.

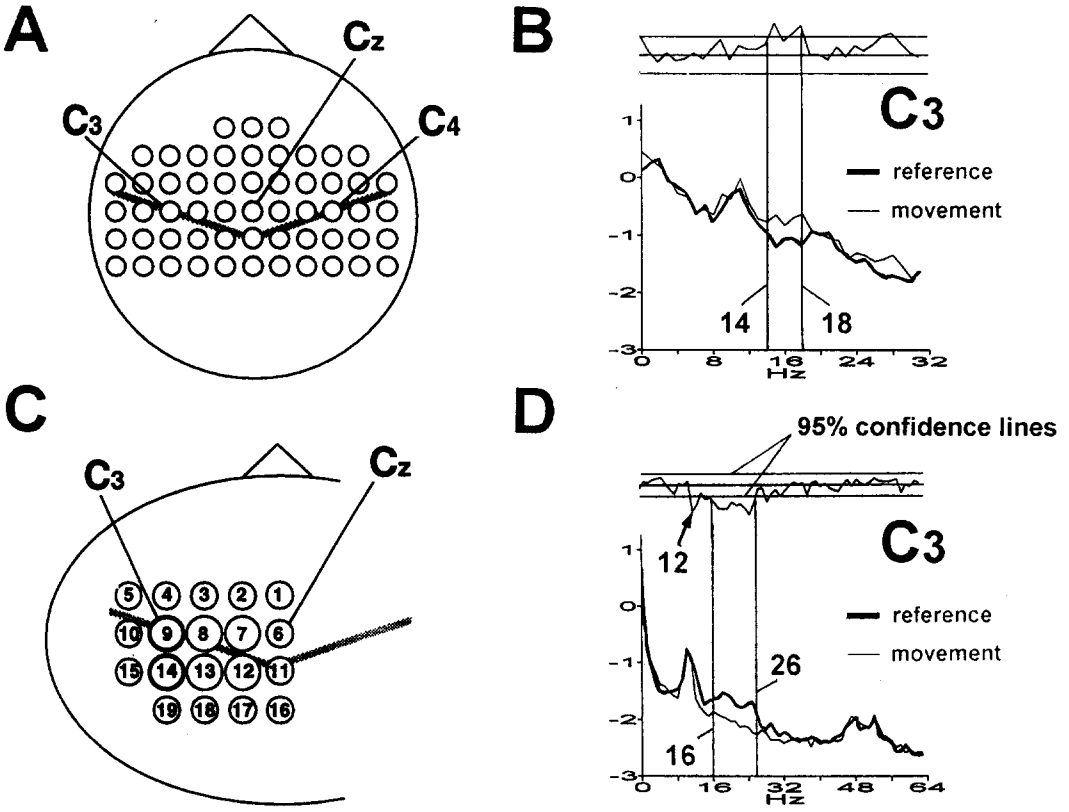


Fig. 1. Electrode placements used in experiment 1 (A, C) and examples for selection of the beta band (B, D). Displayed are 2 logarithmic 1 sec power spectra and their difference with the 95% confidence interval. Full-line spectrum from reference period, thin line spectrum from movement period in D and post-movement period in B. Beta ERS is marked in B and beta ERD in D.

2.1.3. Recordings

EEG was picked up monopolarly from 19 Ag/AgCl electrodes of 7 mm diameter placed 2.5 cm apart in a grid overlying left centroparietal regions. The precise electrode placements used in the study are shown in Fig. 1C. A 56-electrode montage was used in 1 subject (Fig. 1A). EEG was recorded by the BEST system (Fa. Grosseger) with a 0.3 sec time constant and an upper cutoff frequency of 50 Hz. The reference electrode was placed on the right ear. Electrode impedance was kept below 5 kΩ. Surface EMG was recorded from the right hand bipolarly by Ag/AgCl electrodes of 7 mm diameter separated 1.5 cm and placed 5–7 cm above the wrist where the m. extensor digitorum communis was palpated during concurrent index finger extensions. The skin was cleaned with alcohol and massaged gently with Nihon-Kohden Elefix gel. EMG was recorded by a Neurofax (Nihon-Kohden) EEG apparatus. The time constant was 0.1 sec and the upper frequency limit 1000 Hz. EMG was rectified and full-wave integrated by a contour follower (Fridlund, 1979). The time constant of the contour follower was 0.04 sec. A photoelectric sensor was placed on the tip of the index finger to indicate onset and offset of movements. Horizontal EOG was recorded bipolarly by 2 electrodes placed on the skin at the lateral corners of both eyes. Trigger signals, EEG, EOG and integrated EMG were digitized at 128 Hz.

2.1.4. Data processing

For each movement, an 8 sec epoch with 5 sec preceding the onset of the extensor muscle contraction was analyzed. All trials were visually edited for EOG and EMG artifacts. Average trial rejection rate was 20.6%. Reference-free EEG data were obtained by computing the local average reference (Thickbroom et al., 1984) which is similar to the source derivation introduced by Hjorth (1975). Excluding the boundary electrodes, 6 EEG derivations were available. Since electrodes 9 (C3) and 14 (C3p) (see Fig. 1C) showed maximal ERD and ERS, respectively, statistical analyses reported here were carried out for these 2 electrodes only.

Movement-reactive beta components were detected in the 15–26 Hz band in each subject by comparing power spectral densities in the reference interval before (−4.5 to −3.5 sec) and at movement onset (−0.5 to +0.5 sec) from electrode C3. The band showing maximal differences between the reference interval and the activity interval was chosen for ERD/ERS computations. For comparison the movement specific mu frequency band (bandwidth 2 Hz) was also detected by the same procedure as that described for the beta band. An example of 2 superimposed logarithmic 1 sec spectra and their difference are given in Fig. 1D. The beta bands selected are summarized in Table 1. Beta bands ranged from 16.8 (S.D. 1.7) to 22.7 Hz (S.D. 3.1). The EEG was digitally bandpass filtered in the selected bands and power values were computed by firstly squaring the samples and secondly averaging these power samples across trials. To reduce the number of power values, 16 consecutive samples were averaged yielding an 8/sec sampling rate in the resulting power time series. Absolute band power was converted to percentage power using the reference interval as 100%. By convention, an ERD corresponds to a power decrease and an ERS to a power increase (Pfurtscheller, 1992).

Table 1
Beta frequency bands displaying largest beta reactivity during finger movement for experiment 1

Subject	Beta band	Subject	Beta band
S01	16–20 Hz	S08	16–20 Hz
S02	17–27 Hz	S09	15–26 Hz
S04	16–20 Hz	S11	19–21 Hz
S06	18–22 Hz	S12	15–26 Hz
S07	16–20 Hz	S13	20–25 Hz

Data from 10 subjects. Mean frequency band: 16.8–22.7 Hz. Mean frequency: 19.8 Hz.

2.2. Experiment 2

2.2.1. Subjects

Six healthy students (2 females and 4 males, 21–28 years old) participated in this experiment. All subjects were subjective right-handers and were paid for their participation.

2.2.2. Procedure

The subject was seated in a reclining armchair with eyes open, fixing vision on a computer monitor in front of her/him. Her/his forearms were placed on armrests and the hands in a straight resting position on a special platform containing a sensor at the position of each hand. The subject was instructed to make brisk wrist flexion movements (approx. 45°) with the left or right hand using the following behavioral paradigm.

A fixation cross at the center of the monitor indicated the beginning of a trial. Two seconds after trial initiation an arrow was presented at one of the edges of the fixation cross (pointing either left or right). This visual cue indicated the side of movement during this trial, for example, that either left- or right-sided hand movements should be made at any following beep. A series of 4–7 beeps [1] was presented with interstimulus intervals of 3 sec during 1 trial. The subject was asked to respond to each beep by a wrist extension and flexion movement. The direction of the arrow as well as the number of beeps were randomly

[1] The number of beep presentations was randomized between 4 and 7 due to the assumption that in an EEG-based BCI 4–7 EEG patterns have to be classified cumulatively to obtain a high classification accuracy (Pfurtscheller et al., 1996).

selected for each trial and we took brakes of 4 sec between trials.

Each experiment was composed of 3 sessions on different days. In the first 2 sessions (N1, N2) the subject moved her/his wrist without any feedback. These data were used to train a learning classifier with EEG patterns. In the last session (N3) the EEG was classified on-line and used for cursor control. Only the data of the first session were used for statistics.

2.2.3. Recording

EEG was recorded from electrodes C3 and C4 overlying the left and right hand areas (Homan et al., 1987) with a frontal reference electrode. The signals were digitized with 64 Hz. Bandpass filters were set at 5.3 Hz (high pass filter) and 35 Hz (low pass filter).

2.2.4. Data processing

For ERD computation we used the same procedure as for experiment 1. The beta band was determined from two 1 sec spectra, one calculated from the reference period and the other from an interval of −0.5 to 0.5 sec in parallel with the fourth beep. By visual inspection a band of 4 Hz bandwidth showing maximal beta power increase was selected (see Fig. 1B). As reference interval for ERD (%) calculation, a 1 sec period before the presentation of the cue (arrow) was chosen. Event-related EEG epochs from electrodes C3 and C4 ranging from 1 sec before the presentation of the visual cue (arrow) to 1 sec after the fourth beep (resulting in a trial length of 14 sec) were selected for further analysis, so that 3 beep movement sequences could be analyzed for each trial.

2.2.5. Statistical analyses

ERD values in selected time epochs were investigated by the analysis of variance for repeated measures. In experiment 1, the independent variables were bands (mu vs. beta), electrodes (C3 vs. C3p) and periods represented by three 1 sec averages of mu or beta band ERD either preceding or following the onset of extensor muscle contraction. In experiment 2, a 3-way ANOVA for repeated measures included 3 independent variables, i.e., hands (left vs. right), electrodes (C3 vs. C4) and periods. The periods referred to three 1 sec intervals preceding the second, third and fourth beeps. Greenhouse-Geisser ϵ was applied to correct degrees of freedom for the main and interaction effects of periods which was a within-subject variable with more than 2 levels (Vasey and Thayer, 1987). A 95% confidence limit was always employed.

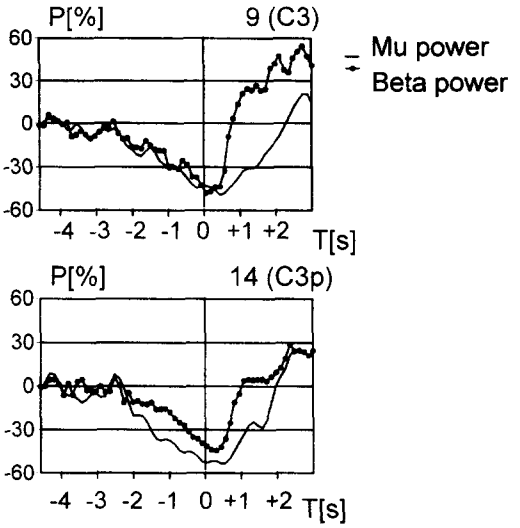


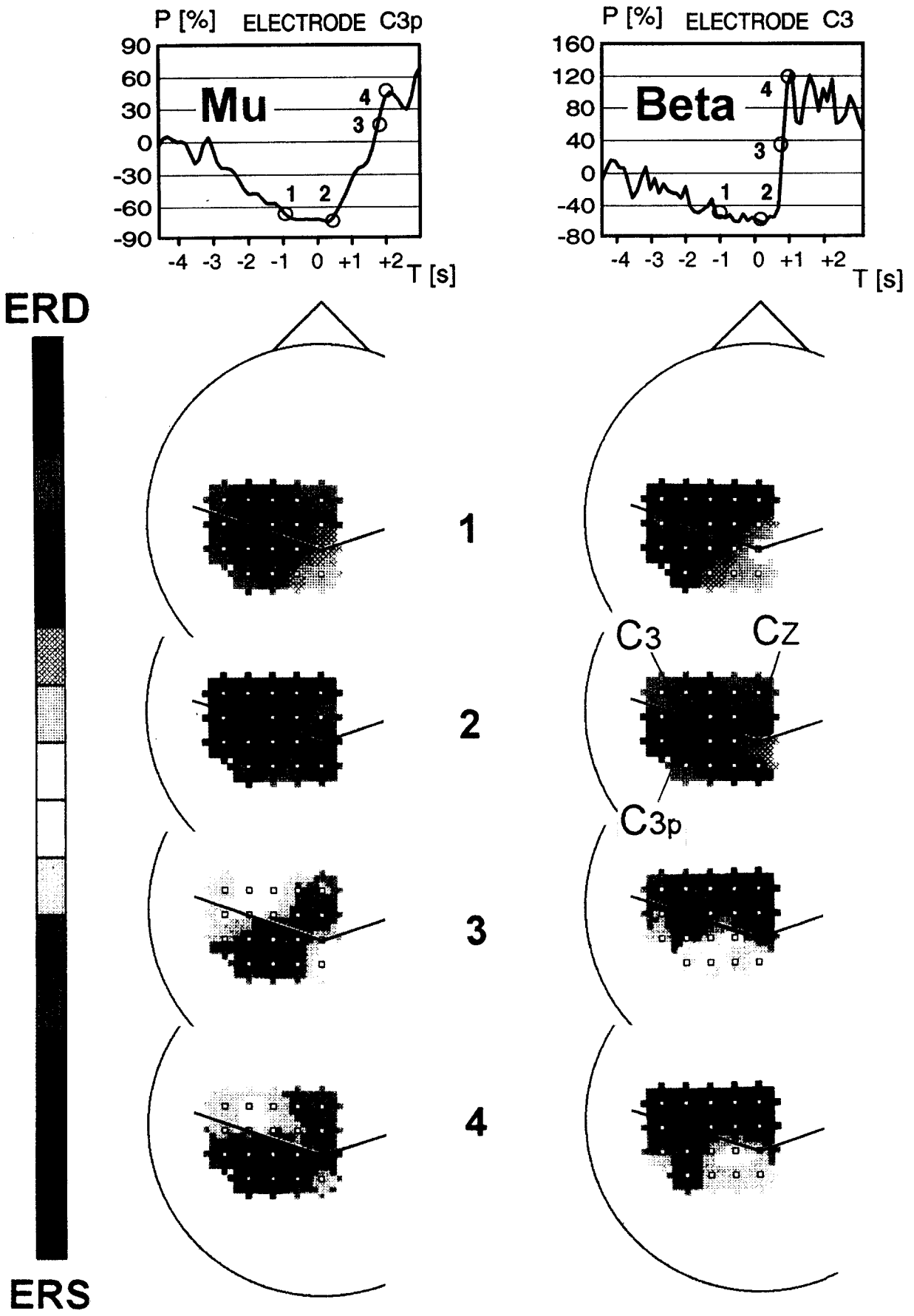
Fig. 2. Grand average beta power time courses (thick line) for 10 subjects during self-paced right index finger movement. Data from 2 electrodes spaced 2.5 cm apart are displayed. Movement onset is at time 0. For comparison, grand average mu power time courses (thin lines) are also displayed.

3. Results

3.1. Beta ERD and ERS in self-paced movements

Grand average band power time courses at electrodes C3 and C3p for the mu and beta bands are displayed in Fig. 2. ERD started 2.5 sec before movement onset in both frequency bands and reached maximal desynchronization shortly after movement onset in both electrodes. The mu rhythm ERD was higher in the preparatory period (−2.0 to −1.0 sec relative to movement onset) compared to the beta band ERD. This finding was documented by ANOVA results (bands (2) × electrodes (2) × periods (3) representing three 1 sec epochs before movement onset). There were significant effects for bands (F (1, 9) = 6.1, P = 0.036), periods (F (2, 18) = 31.5, P = 0.0001, ϵ = 0.819) and the interaction bands × periods (F (2, 18) = 4.1, P = 0.034, ϵ = 0.825) which was largely due to lower power decrease in the beta band in the second second before movement onset (F (1, 9) = 20.3, P = 0.002) but not in the third (F (1, 9) = 0.07, P = 0.80) or in the first (F (1, 9) = 4.24, P = 0.07). There was a tendency toward a

Fig. 3. Topography of beta and mu reactivities in the preparatory period (1), at the point of maximal ERD (2), at the transition to ERS (3), and by maximal ERS (4). Note the postcentral foci of mu and beta ERDs and the precentral focus of beta ERS. Data from subject S09. The scale ranges from +60% to −60%.


P [%] ELECTRODE C3p
90
60
30
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-30
-60
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Mu
1 2 3 4
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160
120
80
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Beta
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higher beta band ERD at electrode C3 which effect attained a borderline significance only (F (2, 18) = 3.48, $P = 0.067$).

The recovery of the central rhythms was faster in the beta band than in the alpha band. Beta band power reached values above the reference power at 0.9 and 1.1 sec after extensor muscle contraction at electrodes C3 and C3p, respectively, and later only at 2.5 sec and 2.0 sec in the mu rhythm band at electrodes C3 and C3p, respectively. In addition, electrode C3 showed more pronounced ERS than electrode C3p. Three-way ANOVA (bands (2), electrodes (2), periods (3)) showed a significant band effect (F (1, 9) = 16.7, $P = 0.003$), period effect (F (2, 18) = 27.5, $P = 0.00001$, $\epsilon = 0.694$) and an interaction bands × electrodes × periods (F (2, 18) = 27.5, $P = 0.00001$, $\epsilon = 0.694$). Further inspection of this interaction showed that it was due to a significant band × electrode effect in the second second after movement onset (F (1, 9) = 25.85, $P = 0.0007$) which in turn was due to higher beta band power at electrode C3 compared to electrode C3p. These results can be summarized as follows: the post-movement beta ERS was dominant at electrode C3.

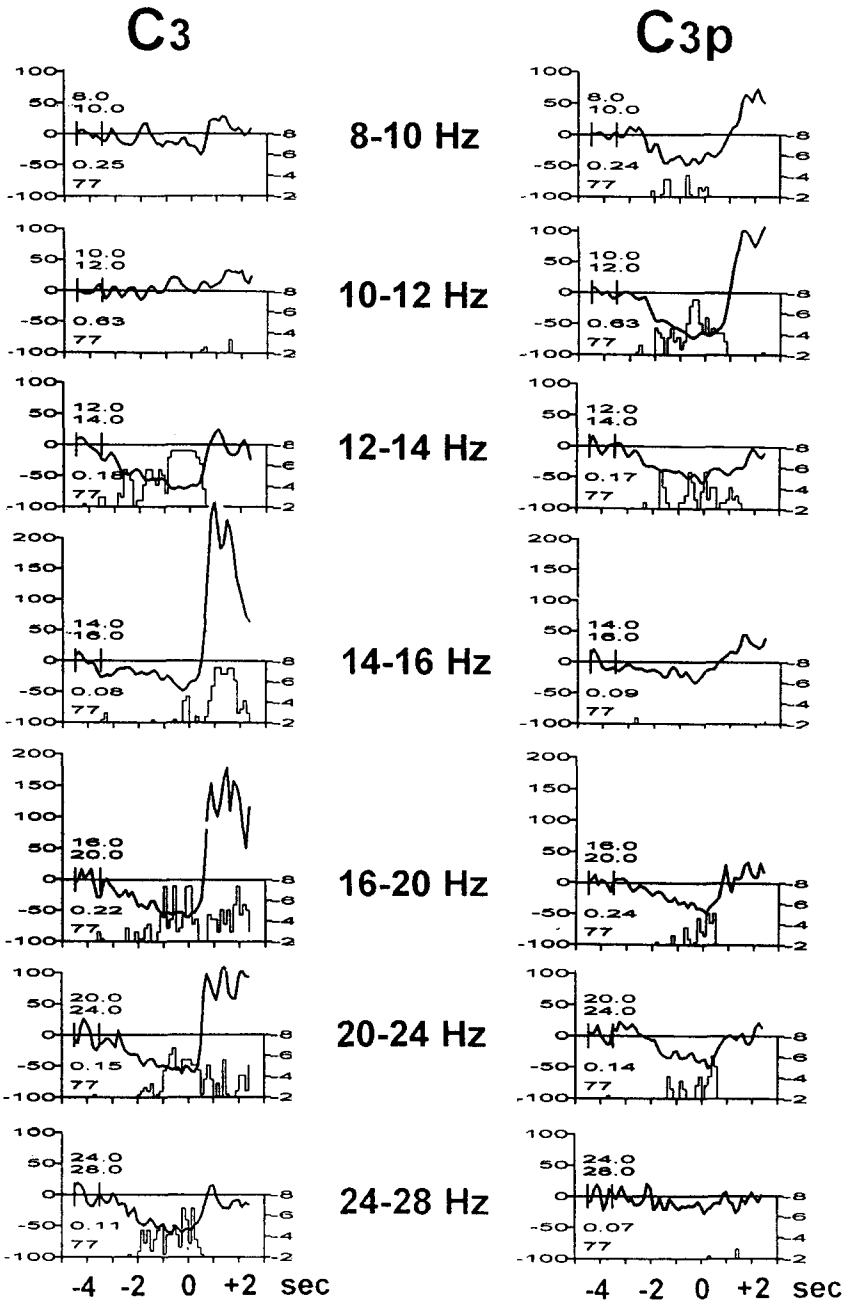

Fig. 4. Band power time courses from 1 subject (S09), 2 electrodes and different frequency bands as indicated. Note the different patterns of power decrease (ERD) and power increase (ERS) in different bands and electrode locations. The analyzed frequency bands are indicated.

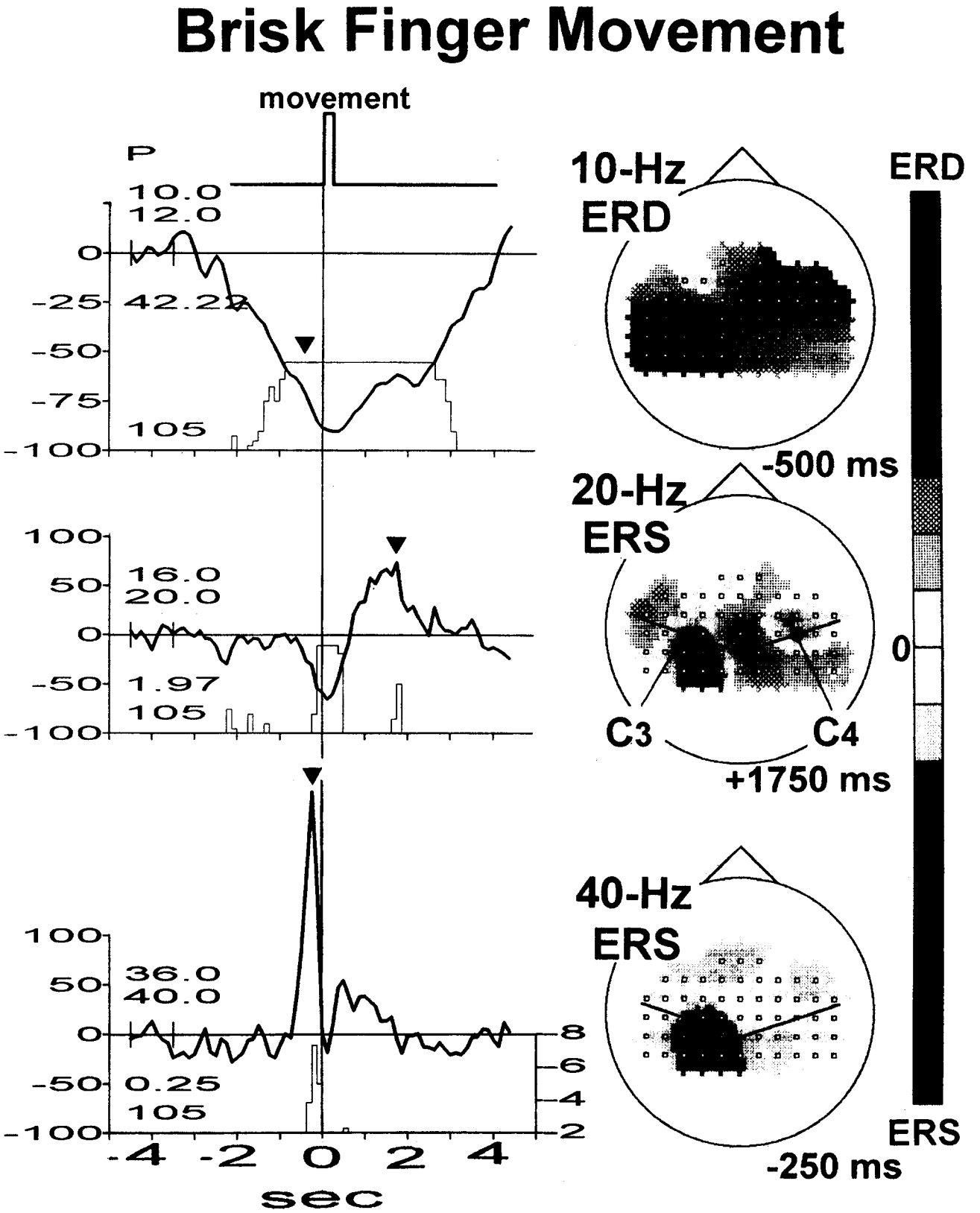

Fig. 5. Band power time courses and topographic maps from 1 subject during brisk, self-paced, right index finger extensions and flexions. Data from 10–12 Hz (mu), 16–20 Hz (lower frequency beta) and 36–40 Hz (gamma) bands are displayed. "Red" indicates event-related power decrease or ERD and "blue" power increase or ERS. The time for calculating the maps is indicated. The color scale is ±80% for the 10 Hz ERD map, ±60% for the 20 Hz ERS map and ±200% for the 40 Hz ERS map. The duration of the movement is indicated.

Fig. 3 gives an illustrative example of the reactivities of mu and central beta rhythms in the form of desynchronization (ERD), recovery and rebound (ERS). The topography of the ERD shows a similar postcentral localization of mu and beta changes both 1 sec before and shortly after movement onset, although mu ERD was more pronounced than beta ERD. The recovery of the beta rhythm including ERS was more prominent at electrode C3 whereas the

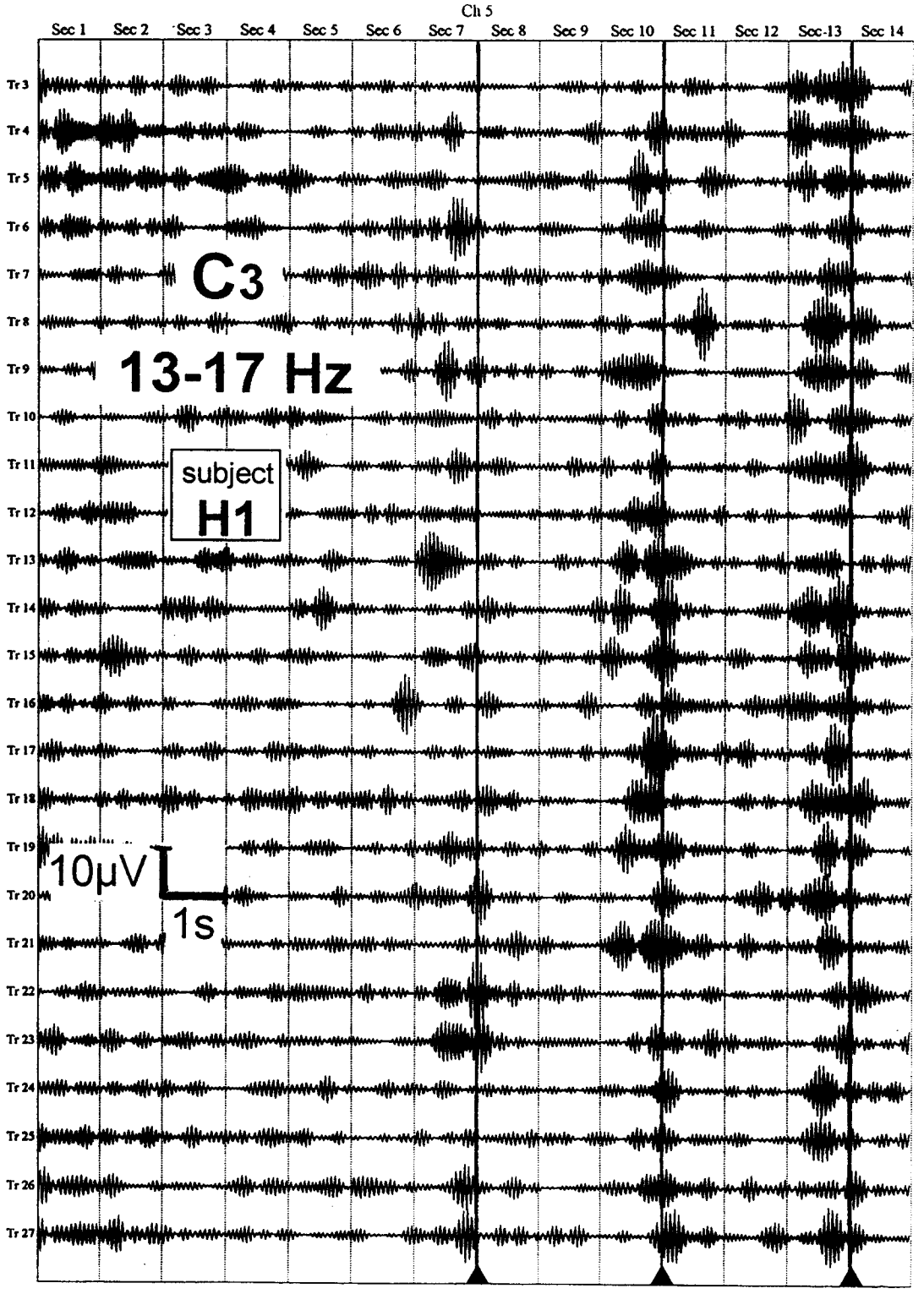

Fig. 6. Examples of single EEG trials (trials 3–27) filtered in the 13–17 Hz band; subject H1. Black triangles mark the time of stimulation.

recovery of mu rhythm had its maximum in the postcentral area (see maps at time points 3 and 4 in Fig. 3).

The importance of frequency band selection and the complex interaction between frequency bands can be seen in Fig. 4. In 1 subject (S09) band power time courses are calculated for electrodes C3 and C3p in the bands 8–10, 10–12, 12–14, 14–16, 16–20 and 20–24 Hz. It can be seen that the 10–12 Hz ERD was located only at electrode C3p and the 24–28 Hz ERD exclusively at electrode C3. The beta ERS was statistically significant in the band 14–16 Hz but was also present with reduced significance in the bands 16–20 and 20–24 Hz. The pre-movement beta ERD was dominant from 16 to 28 Hz. From the differences in mu and beta reactivities at electrodes C3 and C3p especially in the bands 10–12, 14–16 and 24–28 Hz can be speculated that electrode C3 overlies the motor cortex and electrode C3p the somatosensory cortex.

To obtain more detailed information on the topography of the post-movement beta ERS in one of our subjects displaying 40 Hz oscillations prior to movement (Pfurtscheller et al., 1993b, 1994b), a 56-channel EEG study was performed with self-paced, brisk right finger movement. The most important results are summarized in Fig. 5 in the form of time courses and ERD (ERS) maps for the 10–12, 16–20 and 36–40 Hz bands. The movement onset was controlled by EMG recording. A statistically significant short-lasting ERD was followed by a statistically significant ERS in the beta band with a maximum after execution of movements. The post-movement beta ERS was localized at 1 electrode (2.5 cm posterior and 2.5 cm medial to C3). It is of special interest that the "40 Hz oscillations" (40 Hz ERS) prior to movement onset had about the same location as the post-movement 20 Hz beta ERS. Furthermore it has to be noted that the beta ERS was present when the mu rhythm still showed a statistically significant desynchronization.

3.2. Beta ERS during externally paced movements

In Fig. 6 examples of single EEG trial data (filtered in the 13–17 Hz band) of 1 subject recorded from electrode C3 during a beep-paced sequence of right-hand movements are displayed. From these single trial bandpass filtered data it becomes evident that periodic hand movements in intervals of about 3 sec are accompanied by periodic bursts in the beta band.

Table 2
Beta frequency bands displaying largest beta power increase after movement for experiment 2

Subject	Frequency range	Subject	Frequency range
A6	12–16 Hz	D2	16–20 Hz
C7	14–18 Hz	D4	16–20 Hz
D1	14–18 Hz	H1	13–17 Hz

Data from 6 subjects. Mean frequency band: 14.2–18.2 Hz. Mean frequency: 16.2 Hz.

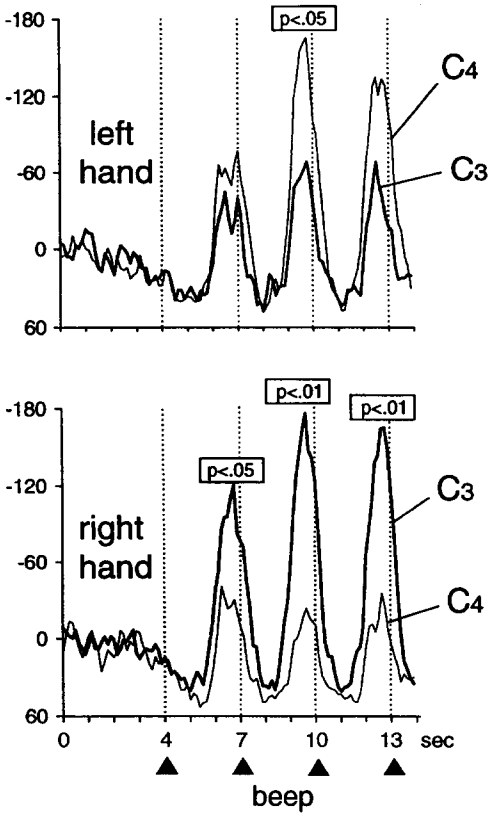

Fig. 7. Grand average band power time courses from 6 subjects with indicated significant hemispheric differences at electrodes C3 and C4 for right- and left-hand movement. The time of stimulation is marked by black triangles.

The grand average band power time courses in individually selected beta frequency bands (see Table 2) of 6 subjects are displayed in Fig. 7. Data from right- and left-hand movements recorded from electrodes C3 and C4 are displayed. It can be seen that after each beep stimulation there was a power decrease followed by a fast recovery and rebound in the form of a short-lasting power increase. This rebound was most dominant over the contralateral hemisphere.

The average ERS values for 2 response sides (left, right), 2 electrode positions (C3, C4) and 3 time periods (representing three 1 sec epochs before presentation of the second, third and fourth beeps) were subjected to an ANOVA for repeated measure factors. A significant main effect due to period (F (2, 10) = 5.8, P = 0.03, ϵ = 0.843) was found, showing a smaller rebound after the first movement compared to both following movements. Further, significant hand × hemisphere (F (1, 5) = 18.9, P = 0.007) and hand × hemisphere × period (F (2, 10) = 10.5, P = 0.01, ϵ = 0.708) interactions were observed. The beta power increase (beta ERS) was larger over the contralat-

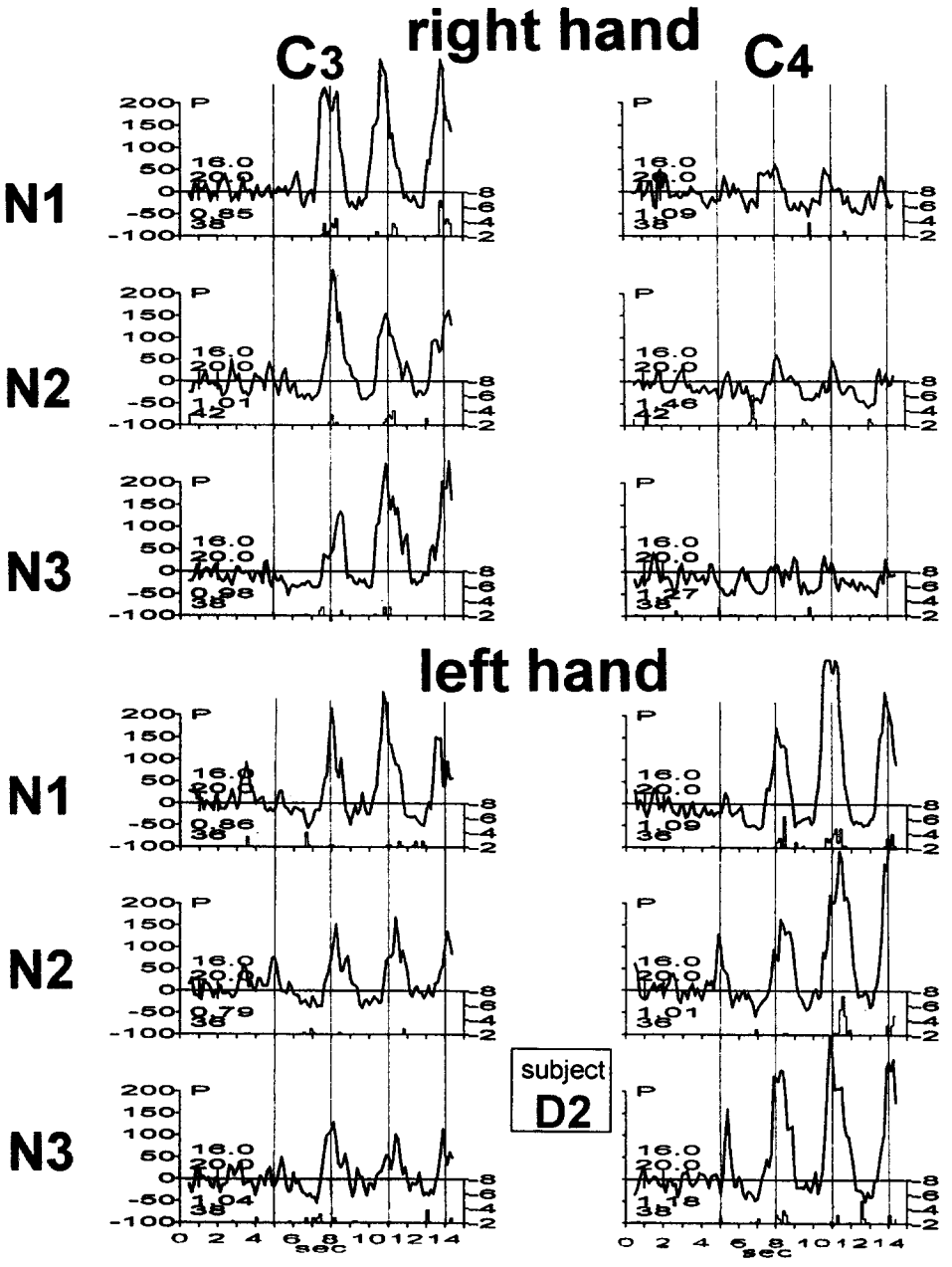

Fig. 8. Examples of beta power (16–20 Hz) time courses with percentage power decrease and increases. The presentation of the beep stimuli is indicated by vertical lines. Data from 3 sessions (N1, N2, N3) on subject D2.

eral compared to the ipsilateral hemisphere. Post hoc analysis using paired t tests (indicated in Fig. 7) revealed a significant higher beta ERS at electrode C3 with right-hand movement in all 3 periods; with left-hand movement the hemispheric differences reached significance only in the second period.

Fig. 8 shows the band power vs. time courses of 1 subject comparing sessions on 3 different days (N1, N2, N3). As can be seen, a clear beta rebound with contralateral dominance was present in all 3 sessions. Like in the grand mean data in Fig. 7 the lateralization was more pronounced with right-hand movement.

4. Discussion

From our results the following conclusions can be drawn.

(i) Finger (wrist) movement is followed by a short-lasting beta synchronization (post-movement beta ERS).

(ii) Post-movement beta ERS is dominant over the

contralateral hemisphere with a focus slightly anterior to the focus of the largest mu desynchronization.

(iii) Post-movement beta ERS can be observed in both internally and externally paced movements.

Self-paced movement can be divided into the phases of planning (programming), execution and recovery. The planning phase starts about 2 sec before movement onset over the contralateral side and is manifested as unilateral desynchronization of mu and central beta rhythms (Chatrian et al., 1959; Pfurtscheller and Berghold, 1989; Derambure et al., 1993). Execution of movement can be monitored by the EMG and results in a bilateral symmetrical ERD. After termination of brisk movement the mu rhythm shows a slow recovery in the period of 2–3 sec following movement, whereas lower beta components recover much faster (within 1 sec). Beta rhythms not only recover to the baseline but also show a transient post-movement power increase characteristic of an event-related synchronization. In the case of voluntary movement we have, therefore, at least 3 characteristic EEG patterns: (i) a contralateral dominant mu and beta ERD prior to movement, (ii) a bilateral ERD during movement, and (iii) a contralateral dominant beta ERS after movement.

It is of interest that the location of the post-movement beta ERS is slightly more anterior compared to the largest mu ERD (see Figs. 2–4). This can be interpreted as follows: the mu rhythm is generated mainly in the somatosensory area and the central beta rhythm (at least some components of it) in the motor area. An investigation of mu and beta rhythms on single trial basis in 3 subjects also revealed a slightly more anterior focus of the beta rhythm compared with the mu rhythm (Pfurtscheller et al., 1994a). MEG measurements made by Salmelin and Hari (1994) revealed a similar result. They interpreted the 10 Hz mu rhythm as located in the primary somatosensory area and the 20 Hz beta rhythm as located in the motor area.

How can we interpret the contralaterally localized post-movement beta ERS? It is speculated that the motor area neurons shift from an activated state during preparation and execution of movement (working cortex) to a resting state after termination of movement (nil-working cortex; Mulholland, 1995) or, in other words, from a processing mode to an "idling" mode.

The term "idling" was first used to describe large amplitude alpha rhythms emerging in the cortex in the absence of a stimulus (Adrian and Matthews, 1934). The idea of cortical idling was also used by Chase and Harper (1971) to describe sensorimotor areas with synchronized 12–14 Hz rhythms. In this sense the occipital alpha rhythm can be considered as "idling" rhythm of the visual system and the mu and central beta rhythms as idling rhythms of the sensorimotor system (Kuhlman, 1978). Oscillations in the alpha and lower beta bands display large amplitudes (synchronization) in such cortical areas not needed to perform a specific task at a certain moment of time (Adrian and Matthews (1934) defined such an area as "an area of the cortex which has nothing to do"). Examples of such areas are the sensorimotor cortex during visual processing (Koshino and Niedermeyer, 1975; Pfurtscheller, 1992) and the visual cortices during motor behavior (Pfurtscheller, 1992; Westphal et al., 1993). A further example is the hand area during foot movement: not only visual processing but also foot movement can result in a localized enhancement of the hand area mu rhythm (Pfurtscheller and Neuper, 1994). Such an enhanced mu rhythm has to be differentiated from a mu rhythm evoked, e.g., by repetitive median nerve stimulation (Narici et al., 1990). In this case it is of interest that the mu rhythm demonstrated prolonged synchronization after termination of stimulation.

An interesting observation is that beta desynchronization is more widespread than synchronization. This becomes especially evident in Fig. 3 with a relatively diffuse pre-movement ERD and a focused post-movement ERS. In this respect it is not astonishing that Toro et al. (1994) as well reported on a widely distributed beta ERD in the 20–30 Hz band. This difference between ERD and ERS is not surprising taking into account that in performing a special task there are always more cortical areas in a processing mode than in an idling mode and therefore EEG desynchronization is dominating over EEG synchronization. In other words, there are always more states of working cortex than of nil-working cortex (Mulholland, 1995).

In the case of externally (beep-) paced movements a bilateral symmetrical ERD during movement was found. After movement, bursts of beta activity (see Fig. 6) occurred over the contralateral side. This beta ERS is not stimulus-locked and belongs, therefore, to the class of induced rhythms or oscillations and cannot be detected by averaging over single EEG trials. It is of interest that this beta ERS was more lateralized in the case of right-hand compared to left-hand movement. Whether this has to do with the right-handedness of the subjects can only be speculated.

Of special interest is the similar spatial location of the post-movement beta ERS and the pre-movement 40 Hz ERS found in 1 subject. In this respect it has to be noted that the subject displaying this 40 Hz ERS was investigated several times (Pfurtscheller et al., 1993b, 1994b) and always showed the same focus of location of these two different types of oscillations. This supports the assumption that not only the 40 Hz ERS, but also the post-movement beta ERS has its origin in or close to the motor cortex. The discrepancy between the findings in our special subject (compare Fig. 5) and the results of the group study (compare Fig. 2) concerning localization of the 20 Hz focus needs further investigation.

It has to be stressed that the post-movement beta ERS around 20 Hz is a quite different phenomenon compared to the ERS around 40 Hz. The 40 Hz ERS is seen as a

correlate of excited neural structures and related, very likely, to sensorimotor integration shortly before or during activation of pyramidal neurons in the motor cortex (Pfurtscheller et al., 1993b). Therefore, oscillations in the beta band can have quite a different physiological meaning depending on the frequency components involved.

There is another aspect which needs some attention. Tables 1 and 2 show that the reactive beta bands found in experiments 1 and 2 are slightly different (experiment 1: mean frequency 19.8 Hz; experiment 2: mean frequency 16.2 Hz). In experiment 1 the beta ERD was determined under the assumption that desynchronization and synchronization are linked phenomena and occupy the same frequency components. In experiment 2 the frequency range of beta power increase was defined. An explanation of these different frequency bands can perhaps be seen in Fig. 4, showing the most dominant beta ERS in the 14–16 Hz band and the most dominant beta ERD in the 16–28 Hz band. This can be interpreted as follows: beta ERD and beta ERS may have different neural substrates. It can be speculated that the relatively widespread beta ERD around 20 Hz is at least partially harmonically related to the mu rhythm (Kuhlman (1978) reported a mean mu frequency of 10.1 Hz), whereas the beta ERS around 16 Hz is perhaps a true response of motor area neurons. In this context it is of interest that the sensorimotor rhythm of Sterman (1977) is defined in the frequency range of 12–15 Hz which is clearly below the first harmonic frequency component of the mu rhythm. To clarify the existence of different beta frequency components and their reactivity in the lower beta band further investigation is needed.

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